

Week	Monday	Wednesday	Friday
Week 1	01/13/25 Course overview and early works of sci-fi Critique of the first sci-fi film Imagining human spaceflight in 1902 Science knowledge survey	01/15/25 Reference frames, acceleration and artificial gravity	01/17/25 What is the nature of space and time? Classical physics and Newton's laws Quiz 1: Intro material (20 pts)
Week 2	01/20/25 No Class: Martin Luther King Jr Day	01/22/25 Newton's laws continued Einstein and modern physics	01/24/25 Einstein and modern physics Relativistic spacetime Problem 1: acceleration
Week 3	01/27/25 Einstein and modern physics Special relativity and time travel	01/29/25 General relativity Black holes and gravitational waves	01/31/25 General relativity continued Gravitational waves and the warp drive Problem 2: time dilation
Week 4	02/03/25 "Future physics": Quantum gravity, worm holes, and the Multiverse Hypothesis Exploration Paper 1: Space & Time	02/05/25 No Class: Spacetime Team Project	02/05/25 No Class: Spacetime Team Project Problem 3: General relativity
Week 5	02/10/25 What is the Universe made of? The standard model of particle physics Spacetime Team Project Due	02/12/25 Dark matter and dark energy Quiz 2: Space and Time (85 pts)	02/14/25 Nuclei, atoms, and gases
Week 6	02/17/25 Liquids, solids, phase transitions	02/19/25 Optical properties; transparent materials	02/21/25 No class today: Optical Team Project Problem 4: matter & materials
Week 7	02/24/25 Metamaterials and invisibility cloaks Optical Team Project due	02/26/25 Chemical energy Midterm Film Critique due	02/28/25 Nuclear weapons; matter & antimatter Exploration Paper 2: Matter & Energy

Week 8	03/03/25 No Class: Fall Break	03/05/25 No Class: Fall Break	03/07/25 No Class: Fall Break
Week 9	03/10/25 Can a machine become self aware? Information storage & data processing	03/12/25 Intelligent robots Quiz 3: Matter and Energy (85 pts)	03/14/25 Bonus class day; Topic TBA
Week 10	03/17/25 The Turing Test	03/19/25 Asimov's Laws and robot behavior	03/21/25 Artificial conciousness & robot ethics Exploration Paper 3: AI
Week 11	03/24/25 Are we alone in the Universe? Conditions necessary for life to exist Quiz 4: AI (50 pts)	03/26/25 Finding habitable planets	03/28/25 The science behind the SETI project Exploration Paper 4: Extraterrestrial Intelligence
Week 12	03/31/25 What does it mean to be human? Insight from Frankenstein Medical technology Quiz 5: Extraterrestrial Intelligence (30 pts)	04/02/25 Cell structure and radiation damage DNA, genome sequencing & gene editing	04/04/25 Human teleportation
Week 13	04/07/25 What can we learn from an android about what it means to be human? Exploration Paper 5: Being Human	04/09/25 How can we solve our problems Public perception of science The scientific method	04/11/25 Misplaced faith in science Misunderstanding of science Quiz 6: Being Human (50 pts)
Week 14	04/14/25 Misuse of science in advertising Final Film Critique due	04/16/25 No Class: Thanksgiving Break	04/18/25 No Class: Thanksgiving Break
Week 15	04/21/25 What lies ahead? Examples of sci-fi predicting the future How sci-fi storytelling changes with time	04/23/25 Imagining the future Quiz 7: Problem solving (30 pts)	04/25/25 Last day of class – closing discussion